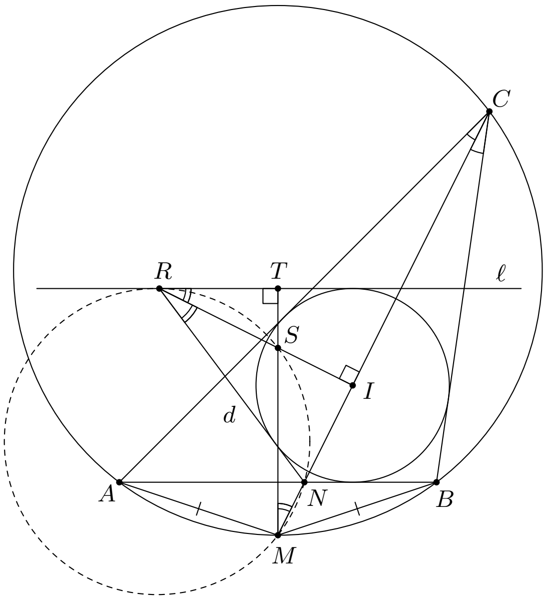
**Problem 105.** Let $\triangle ABC$ be scalene with incentre $I$ and with $CN$ as the angle bisector of angle $C$. The line $CN$ meets the circumcircle of $\triangle ABC$ again at $M$. The line $\ell$ is parallel to $AB$ and touches the incircle of $\triangle ABC$. The point $R$ on $\ell$ is such that $CI \perp IR$. The circumcircle of $\triangle MNR$ meets the line $IR$ again at $S$.

Prove that $AS = BS$.

**Solution.**




Below $\angle(n, m)$ denotes the directed angle between lines $n$ and $m$.

Let $d$ be the tangent to the circumcircle of $\triangle ABC$ containing $N$ and different from AB. Then $\angle(\ell, CI) = \angle(NB, NI) = \angle(NI, d)$. Since $CI \perp IR$, the line $d$ contains R because of symmetry with respect to $IR$.

Let $T = MS \cap \ell$. We have

$$\angle(MN, MS) = \angle(RN, RS) = \angle(RS, RT),$$

that is, $R, T, I, M$ are concyclic. Hence

$$\angle(RT, MT) = \angle(RI, MI) = 90^\circ.$$

It follows that $MS \perp AB$. But $M$ belongs to the perpendicular bisector of $AB$, and so does $S$. Therefore $AS = BS$. $\square$

**Problem 106.** Find all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ such that

$$f(4x + 3y) = f(3x + y) + f(x + 2y) \quad \text{for all } x, y \in \mathbb{Z}. \tag{1}$$

**Solution.** The answer is

$$f(x) = \begin{cases} \dfrac{ax}{5} & \text{if } 5 \mid x, \\ bx & \text{otherwise,} \end{cases}$$

where $a, b$ are arbitrary integers.

Putting $x = 0$ in $(1)$, we obtain

$$f(3y) = f(y) + f(2y) \tag{2}$$

Next, putting $y = -2x$ in $(1)$ gives $f(-2x) = f(x) + f(-3x) = f(x) + f(-x) + f(-2x)$ (in view of (2)). Hence

$$f(-x) = -f(x) \tag{3}$$

Now, putting $x = 2z - v, y = 3v - z$ in $(1)$, we obtain

$$f(5z + 5v) = f(5z) + f(5v) \tag{4}$$

for all $z, v \in \mathbb{Z}$. It follows from Cauchy equation that $f(5t) = tf(5)$ for $t \in \mathbb{Z}$, or equivalently, $f(x) = \frac{ax}{5}$ for any $x$ divisible by 5, where $a = f(5)$.
Now we claim that

$$f(x) = bx \tag{5}$$

where $b = f(1)$, for all $x$ not divisible by 5. In view of (3) it suffices to prove the claim for $x > 0$.

We use induction in $k$ where $x = 5k + r$, $k \in \mathbb{Z}$, $0 < r < 5$. For $x = 1$, (5) is obvious. Putting $x = 1$, $y = -1$ in (1) gives $f(1) = f(2) + f(-1)$, whence $f(2) = f(1) - f(-1) = 2f(1) = 2b$. Then $f(3) = f(1) + f(2) = 3b$ by (2). Finally, (1) with $x = 1, y = 0$ gives $f(4) = f(3) + f(1) = 3b + b = 4b$. These prove base case.

Now suppose (5) is true for $x < 5k$. Observe that

- $$\begin{aligned} f(5k+1) &= f(4(2k-2) + 3(3-k)) \\ &= f(3(2k-2) + (3-k)) + f((2k-2) + 2(3-k)) \\ &= f(5k-3) + f(4) \\ &= (5k-3)b + 4b \\ &= (5k+1)b; \end{aligned}$$
- $$\begin{aligned} f(5k+2) &= f(4(2k-1) + 3(2-k)) \\ &= f(3(2k-1) + (2-k)) + f((2k-1) + 2(2-k)) \\ &= f(5k-1) + f(3) \\ &= (5k-1)b + 3b \\ &= (5k+2)b; \end{aligned}$$
- $$\begin{aligned} f(5k+3) &= f(4 \cdot 2k + 3(1-k)) \\ &= f(3 \cdot 2k + (1-k)) + f(2k + 2(1-k)) \\ &= f(5k+1) + f(2) \\ &= (5k+1)b + 2b \\ &= (5k+3)b; \end{aligned}$$
- $$\begin{aligned} f(5k+4) &= f(4(2k+1) + 3(-k)) \\ &= f(3(2k+1) + (-k)) + f((2k+1) + 2(-k)) \\ &= f(5k+3) + f(1) \\ &= (5k+3)b + b \\ &= (5k+4)b. \end{aligned}$$

This completes the induction.

It remains to check that the function found above satisfies (1). For this it is enough to note that 5 either divides all the numbers $4x + 3y, 3x + y, x + 2y$ or it does not divide any of these numbers because

$$3x + y = 5(x + y) - 2(x + 2y) = 2(4x + 3y) - 5(x + y). \qquad \square$$

**Remark.** Note that the inductive step can be much simplified. We only need the following observations:

- Every integer $n > 3 \times 4 - 3 - 4 = 5$ can be written as $4x + 3y$ for nonnegative integers $x, y$.
- $4x + 3y > 3x + y, x + 2y$.
- $5 \mid 4x + 3y$ if and only if $5 \mid 3x + y$ if and only if $5 \mid x + 2y$.

**Problem 107.** Find the maximum possible value of $k$ for which in a 20-element set there are $2k + 1$ different 7-element subsets such that each of these subsets intersects exactly $k$ other subsets.

**Solution.** The answer is $k = 2$.

Let $M$ be the set of residues modulo 20. An example is given by the sets

$$A_i = \{4i+1, 4i+2, 4i+3, 4i+4, 4i+5, 4i+6, 4i+7\} \subseteq M, \quad i = 0, 1, 2, 3, 4.$$

Let $k \geqslant 2$. Obviously among any three 7-element subsets there are two intersecting subsets. Let $A$ be any of the $2k + 1$ subsets. It intersects $k$ other subsets, say $B_1, \ldots, B_k$. The remaining subsets, say $C_1, \ldots, C_k$ do not intersect $A$ and are therefore pairwise intersecting. Since each $C_i$ intersects $k$ other subsets, it intersects exactly one $B_j$ . This $B_j$ cannot be the same for all $C_i$ because $B_j$ cannot intersect $k + 1$ subsets. Thus there are two different $C_i$'s intersecting different $B_j$'s. Let $C_1$ intersect $B_1$ and $C_2$ intersect $B_2$. All the subsets that do not intersect $C_1$ must intersect each other; there is an $A$ among them, therefore they are $A$ and all $B_i$'s, $i \neq 1$. Hence every $B_i$ and $B_j$, $i \neq 1, j \neq 1$, intersect. Applying the same argument to $C_2$ we see that any $B_i$ and $B_j$, $i \neq 2, j \neq 2$, intersect. We see that the family $A, B_1, \ldots, B_k$ contains only one pair, $B_1$ and $B_2$, of non-intersecting subsets, while $B_1$ intersects $C_1$ and $B_2$ intersects $C_2$. For each $i$ this list contains $k$ subsets intersecting $B_i$. It follows that no $C_i$ with $i > 2$ intersects any $B_j$, that is, there are no such $C_i$, and $k \leqslant 2$. $\square$

**Problem 108.** Let $p$ be a prime number and $a, k$ be positive integers such that $p^a < k < 2p^a$.
Prove that there exists a positive integer $n$ such that

$$n < p^{2a} \quad \text{and} \quad \binom{n}{k} \equiv n \equiv k \pmod{p^a}.$$

**Solution.** First of all, we will prove that

$$\binom{k+tp^a}{k}, \quad t = -1, 0, 1, \ldots, p^a - 2$$

is a complete residue system modulo $p^a$.
Let $v_p(m)$ be the highest power of $p$ in the prime factorization of $m$, and let $r_p(m) = \dfrac{m}{p^{v_p(m)}}$.
Since $p^a < k < 2p^a$, and so for $t \in \{-1, 0, 1, \ldots, p^a - 2\}$, we have

$$\binom{k+tp^a}{k} = \prod_{i=1}^{k} \frac{i+tp^a}{i} = \frac{p^a+tp^a}{p^a} \prod_{\substack{i=1\\ i\neq p^a}}^{k} \frac{i+tp^a}{i} = (1+t) \prod_{\substack{i=1\\ i\neq p^a}}^{k} \frac{r_p(i)+tp^{a-v_p(i)}}{r_p(i)} = (1+t)\frac{M(t)}{M},$$

where $M(t) = \displaystyle\prod_{\substack{i=1\\ i\neq p^a}}^{k} \left(r_p(i) + tp^{a-v_p(i)}\right)$ and $M = \displaystyle\prod_{\substack{i=1\\ i\neq p^a}}^{k} r_p(i)$.
For $1 \leqslant i \leqslant k$ with $i \neq p^a$, since $v_p(i) \leqslant a-1$, we have

$$\gcd\left(r_p(i) + tp^{a-v_p(i)}, p\right) = \gcd(r_p(i), p) = 1$$

and so $\gcd(M(t), p) = \gcd(M, p) = 1$.
Now suppose that there are integers $t, s$ with $-1 \leqslant t < s \leqslant p^a - 2$ such that

$$\binom{k+tp^a}{k} \equiv \binom{k+sp^a}{k} \pmod{p^a}.$$

Then

$$(1+t)M(t) = M\binom{k+tp^a}{k} \equiv M\binom{k+sp^a}{k} = (1+s)M(s) \pmod{p^a}.$$

Now let $s - t = p^b\ell$, where $b = v_p(s-t)$ and $\ell = r_p(s-t)$. Then $b \leqslant a-1$ and $\gcd(\ell, p) = 1$. Hence

$$\begin{aligned}
(1+t)M(t) &\equiv (1+s)M(s) \\
&= \left(1+t+p^b\ell\right) M(s) \\
&= p^b\ell M(s) + (1+t) \prod_{\substack{i=1\\ i\neq p^a}}^{k} \left(r_p(i) + \left(t+p^b\ell\right) p^{a-v_p(i)}\right) \\
&= p^b\ell M(s) + (1+t) \prod_{\substack{i=1\\ i\neq p^a}}^{k} \left(r_p(i) + tp^{a-v_p(i)} + \ell p^{a-1-v_p(i)} p^{b+1}\right) \\
&\equiv p^b\ell M(s) + (1+t)M(t) \pmod{p^{b+1}},
\end{aligned}$$

and so $p^b\ell M(s) \equiv 0 \pmod{p^{b+1}}$. But this contradicts the fact that $\gcd(\ell, p) = \gcd(M(s), p) = 1$ and so, for any $t, s$ with $-1 \leqslant t < s \leqslant p^a - 2$, we have

$$\binom{k+tp^a}{k} \not\equiv \binom{k+sp^a}{k} \pmod{p^a},$$

which means that $\binom{k+tp^a}{k}, t = -1, 0, 1, \ldots, p^a - 2$ is a complete residue system modulo $p^a$.

Now take $t \in \{-1, 0, 1, \ldots, p^a - 2\}$ such that $\binom{k+tp^a}{k} \equiv k \pmod{p^a}$. Moreover, take $n = k + tp^a$ and so $0 < n < 2p^a + (p^a - 2)p^a = p^{2a}$ and $\binom{n}{k} \equiv n \equiv k \pmod{p^a}$. This completes the proof. $\square$